To Virtualize or Not to Virtualize:

Experiences from Building Two Generations of Virtualized AI Infrastructure for LLM Training

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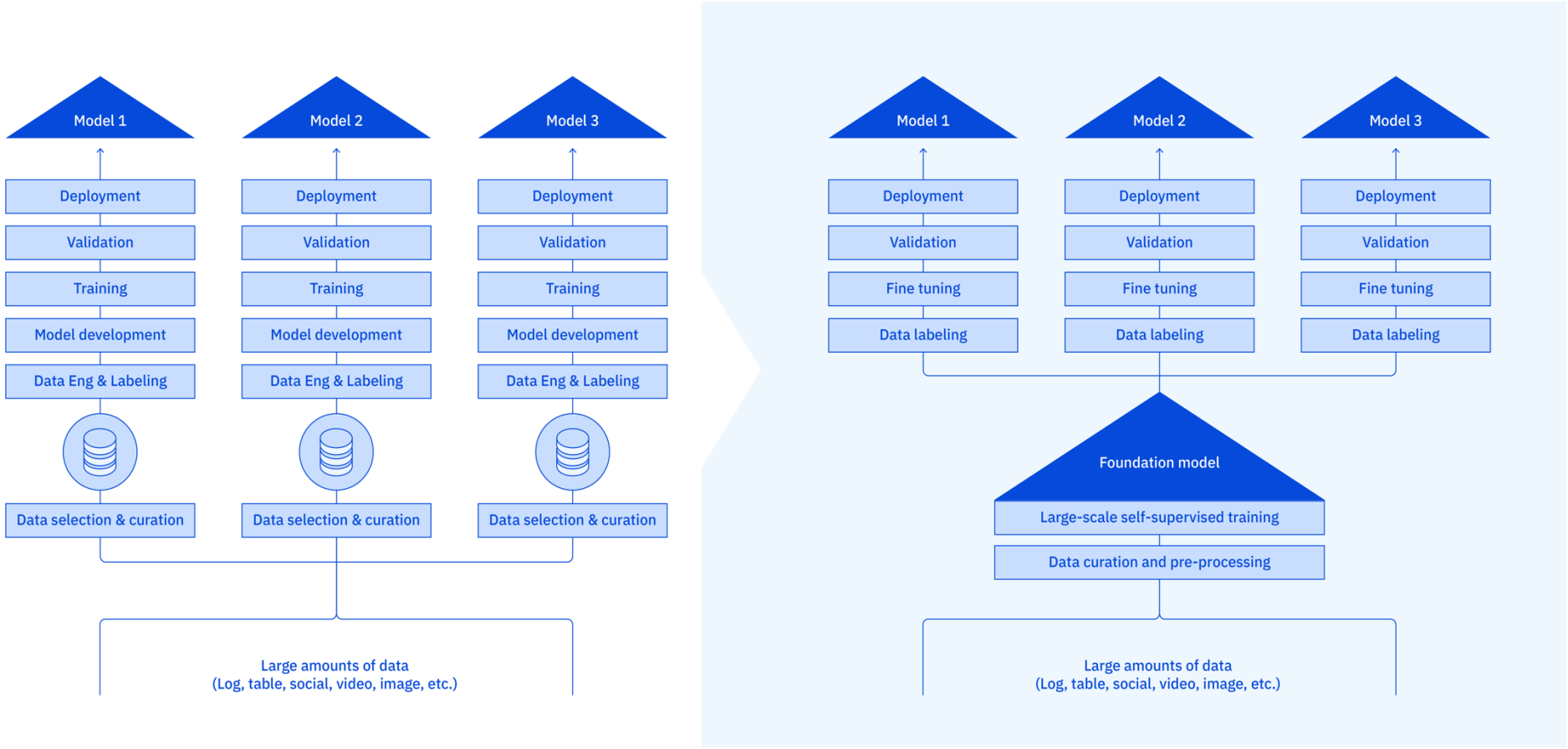
Background

AI Trend Change since 2020/2021: Foundation Models



Background

Focus on Foundation Model Pre-Training

Data Preparation

Workflow of steps (e.g., remove hate and profanity, deduplicate)

Hours to days

10-2000+ low to mid-end CPU cores

10+ low to mid-end GPUs per

10-100+ concurrent jobs

on-prem | Public clouds

Pre-Training

Long-running job on massive infrastructure

weeks to months

10-500+ high-end GPUs (per job)

10+ concurrent jobs

on-prem | Public clouds

Fine-Tuning

Model tuning with custom data set for downstream tasks

minutes to hours

1-64+ mid to high-end GPUs (per job)

100+ concurrent jobs

on-prem | Public clouds

Inference

May have sensitivity to latency, throughput, power

sub-second API request

1-16+ GPUs per fine tuning task

Fraction to multiple GPUs per inference, or specialized accelerator

Thousands of API requests

on-prem | Public clouds | Edge

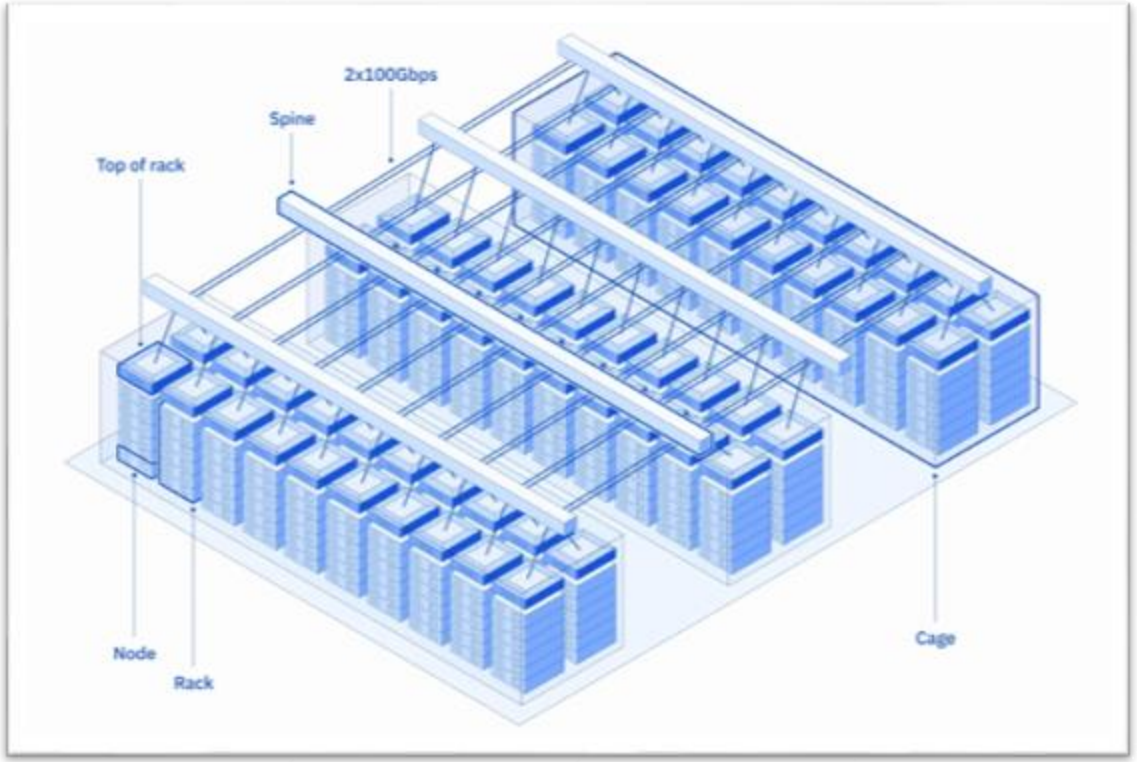
System Requirements for Pre-Training

High-Performance, Elasticity, Power Efficiency , Operational Agility

- Pre-training workload behavior similar to High Performance Computing workloads:
  - GEMM calculations that can benefit from parallel programming paradigm.

- IBM Internal Cluster:
  - Used by AI researchers.
  - Demand for Multi-tenancy and Cloud-native.

"Can we build a cloud (virtualization-based) system for LLM training workloads with off-the-shelf hardware and a primarily open-source system stack while delivering HPC-like performance?"

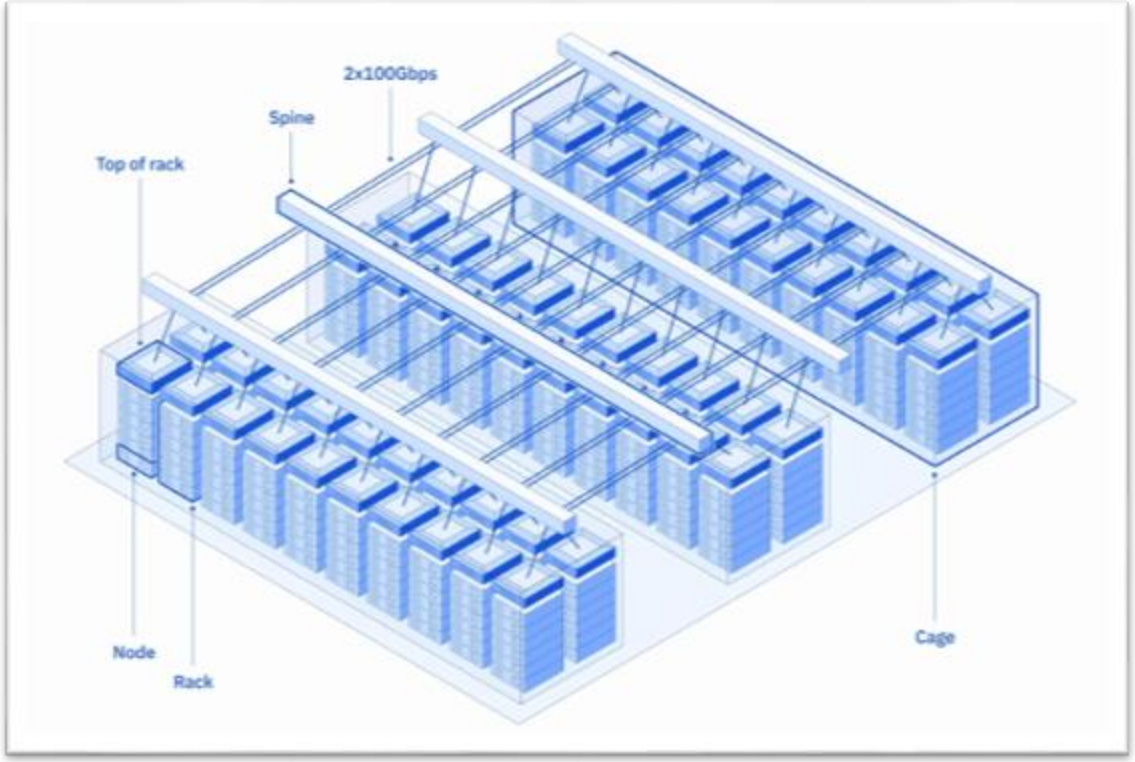


Vela: Cloud-Native AI System Architecture

Overview: VM-based IaaS Cloud System

Compute, Network, and Storage:

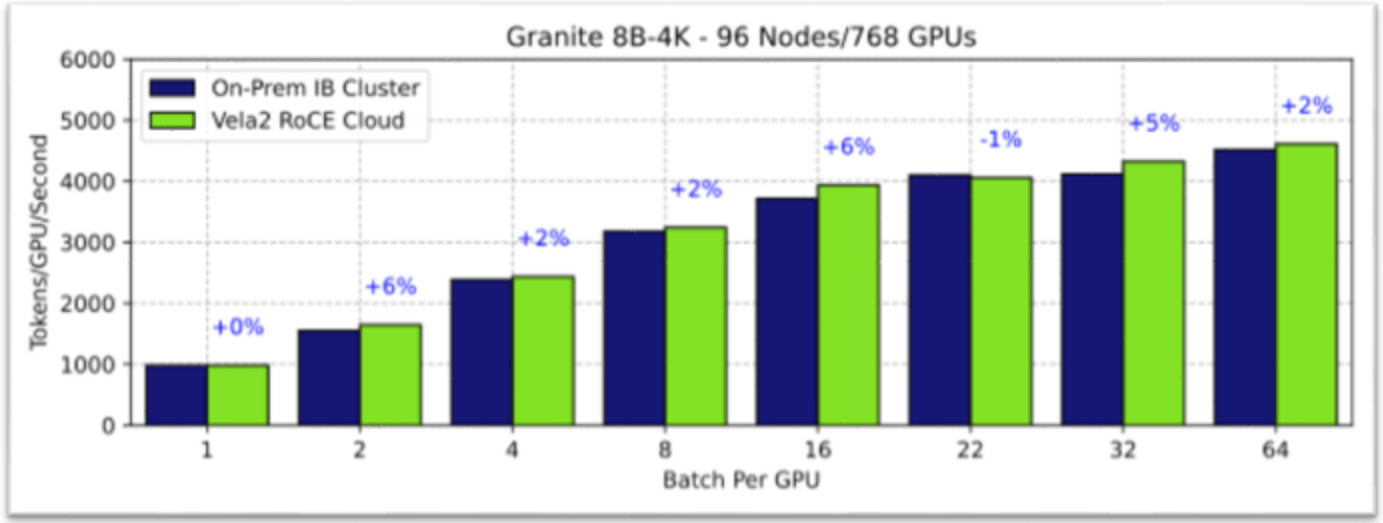
- Off-the-Shelf Hardware
- Virtual Machines: Linux, KVM, QEMU
- RDMA over Converged Ethernet (RoCE)
- GPUs: Full-Passthrough Mode
- RoCE-capable SRIOV-based vNICs
- NUMA and PCIe Topology Aware
- P2P DMA: GPU-NIC (GDR: GPU Direct RoCE)
- Vendor-Agnostic Software Defined Networking (SDN)
- Congestion Control: ECMP, RDMA QP, Flow Control
- Three-tier Storage Solution: NVMe, GPFS, COS



Performance Evaluation in Production

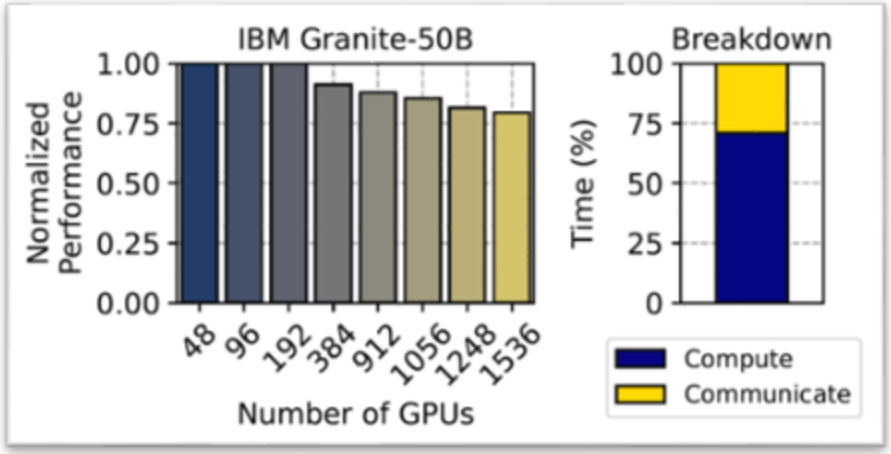
At-Scale Training Performance with VMs Similar to Bare-Metal

Bare-metal InfiniBand vs Virtualized RoCE



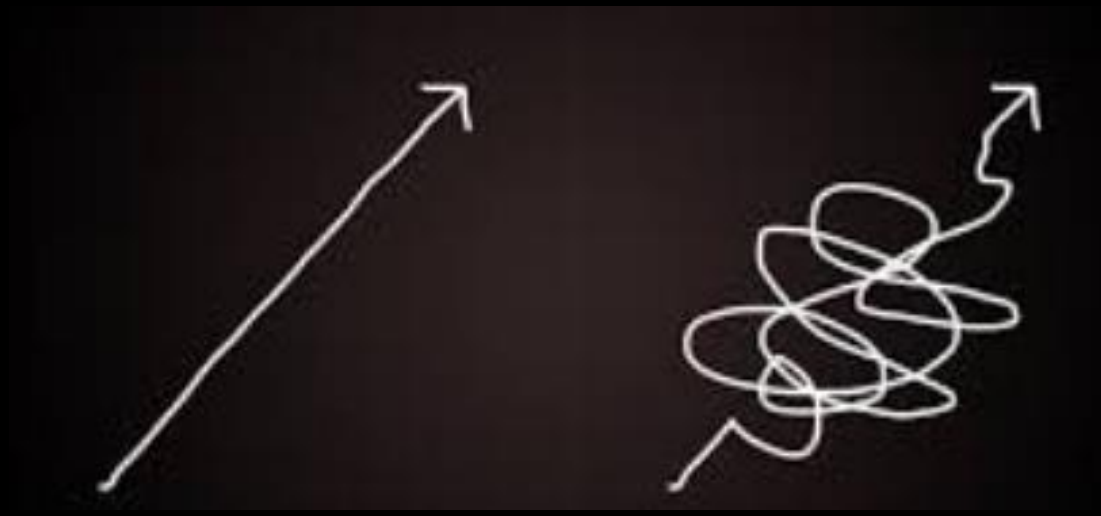
Vela Gen. 2: NVIDIA H100

Bare-metal RoCE vs Virtualized RoCE



Vela Gen. 1: Nvidia A100

The *"Compute-Side"* Journey: From Lab to Production



Expectation vs Reality

Key Challenges: Virtualizing Modern AI Hardware

Expectation Mismatch: Virtualization Stack and Cross-Vendor Stack

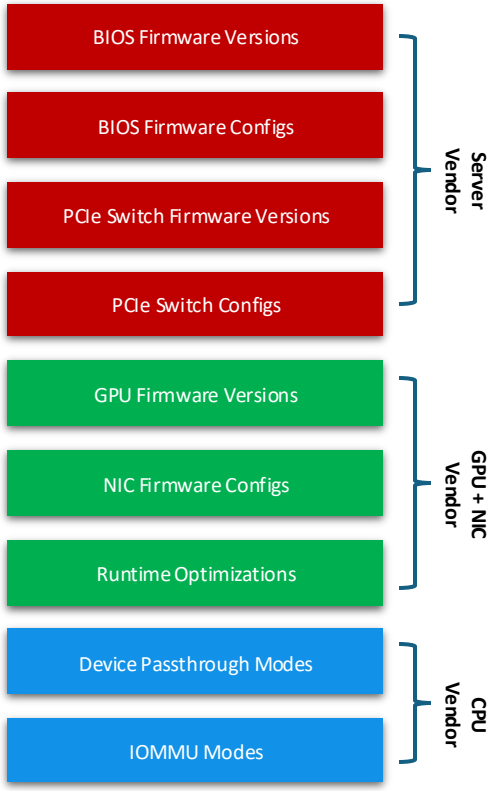
Traditional virtualization was never designed for:

- Multi-hundred-GB/s GPU fabrics (NVLink/NVSwitch)
- PCIe peer-to-peer DMA
- RDMA-capable NICs

Most OEMs do not certify their AI/HPC SKU's for Linux+KVM+QEMU-based virtualization stack.

- Out-of-the-Box Performance for AI Workloads is suboptimal.

- Achieving near–bare-metal performance in VMs requires deep co-design across BIOS, NIC firmware, hypervisor, and guest OS.



H/W Configuration Space

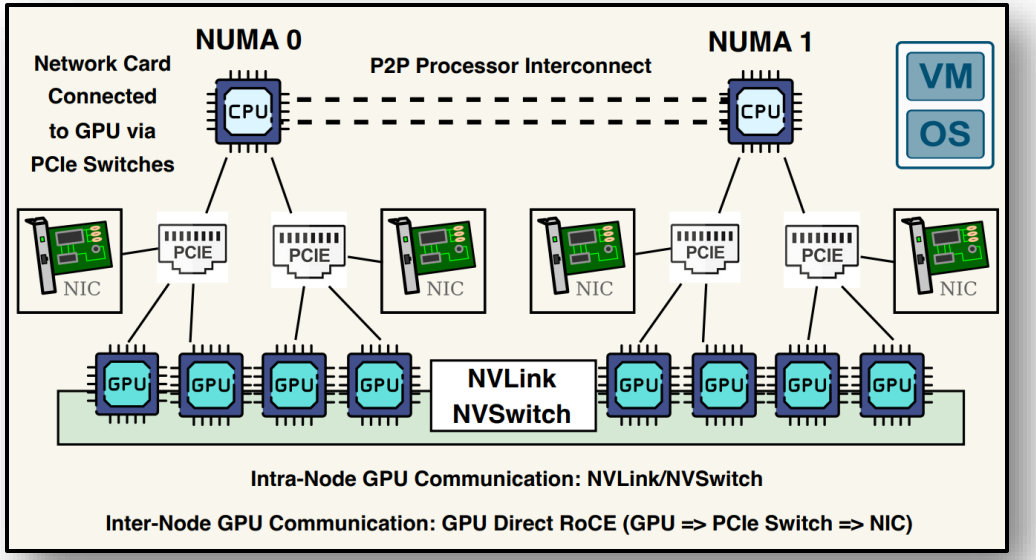
Key Challenges: Virtualizing Modern AI Hardware

Complex Server Design: AI Servers Pack Many Components

- Hierarchical PCIe Topology
- Intra-node Fabrics (e.g., NVLink)
- Multiple PCIe Root Complex
- Multiple NUMA Domains

Hardware/Servers designed for AI/HPC workloads are highly optimized but primarily for bare-metal deployments.

Virtualization: Performance vs Security
- IOMMU Security Requirements Kicks in.
- PCIe Peer-to-Peer DMA gets impacted; key for GPU Direct RDMA, GPU Direct Storage.
- Not all devices implement PCIe ATC/ATS.



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Many Vendors Still Lack PCIe ACS/ATS support.

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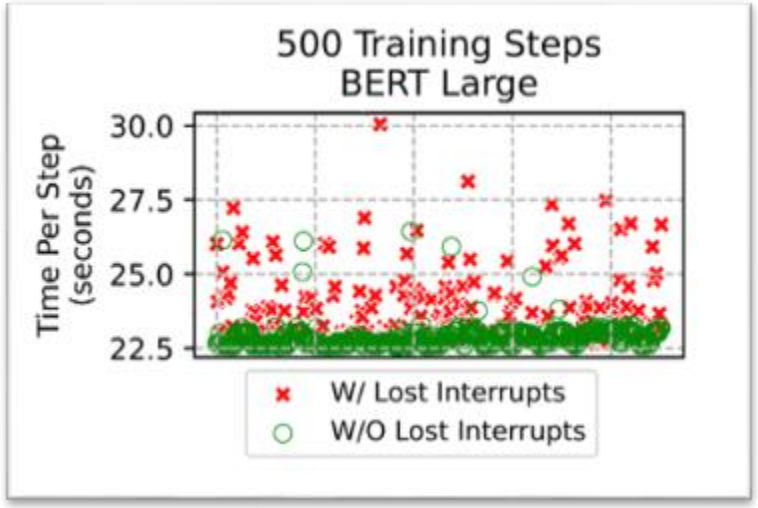
Key Challenges: Virtualizing Modern AI Hardware

Linux Kernel (as a hypervisor) needs to be “battle tested” for AI/HPC Systems

Identified issues:

- Race condition in Kernel:
  - Led to intermittent performance drop when training using SRIOV + TCP.
- Mis-configuration of Kernel data structures:
  - Led to PCIe configurations (e.g., MaxReadReq, ATS) being reset on every Guest OS reboot when IOMMU in passthrough mode.
- Potential RAS Issue due to VFIO Interface:
  - Privileged access to PCIe device extended registers in Guest – can lead to hypervisor crash from the guest OS.



Conclusion: Virtualization is Powerful, but…

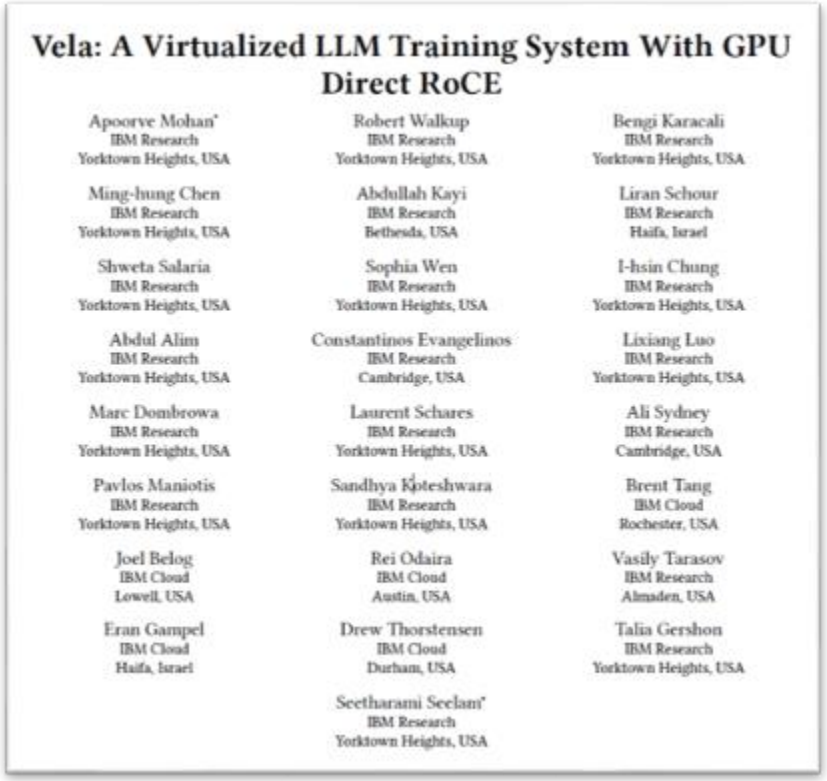
From the Ghost of Christmas Past…

Warning: Proceed With CAUTION!!!

- Delivering state-of-the art performance with open-source virtualization stack (Linux, KVM, QEMU) is possible but “non-trivial”.
- You may have the best-in-class hardware, but runtime mis-configurations can/will bring the performance down.

Questions?

ACM ASPLOS'25: https://dl.acm.org/doi/abs/10.1145/3676641.3716280



Vela: A Virtualized LLM Training System With GPU Direct RoCE

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